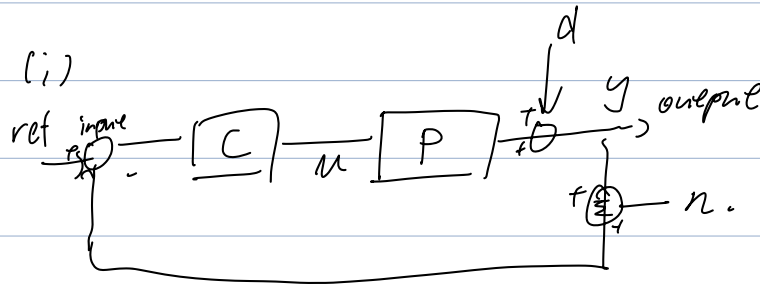


2021.

Q3.

(a). (i)

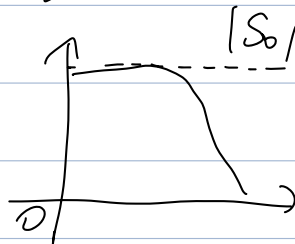
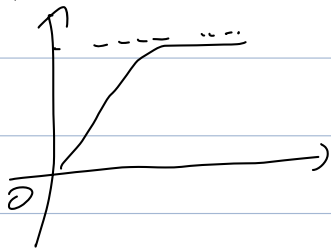


$$[r - (y + n)]CP + d = y$$

$$\Rightarrow y = \frac{(r - n)CP + d}{1 + CP}$$

$$= \frac{CP}{1 + CP} \cdot r + \frac{1}{1 + CP} \cdot d - \frac{CP}{1 + CP} \cdot n$$

$$[T_o] = T_o r + S_o d - T_o n$$



the vector margin: $S_m = \frac{1}{\max |1 + L_o(j\omega)|} = \frac{1}{\max |S_o(j\omega)|}$

inverse: $T_m = \frac{1}{\max |1 + \frac{1}{L_o(j\omega)}|} = \frac{1}{\max |T_o(j\omega)|}$

peaking of S_o and T_o must be avoided not

only to achieve robust stability,

but also to ensure acceptable close-loop time-domain response.

$$(iii), \quad \frac{\partial S_o}{S_o} = S \frac{\partial P}{P} \Rightarrow S_P = \frac{P}{S_o} \cdot \frac{\partial S_o}{\partial P}$$

$$S_o = \frac{1}{1+CP} \quad S_P = P(1+CP) \cdot \frac{-C}{(1+CP)^2}$$

$$= -\frac{CP}{1+CP}$$

b. Loop-shaping is a controller design method in which the controller $C(s)$ is chosen so that the open-loop transfer function $L(s) = C(s)P(s)$ has a desirable frequency-domain shape, given $P(s)$.

c. Varying controller gain simply changes the mag., not phase. it can $\uparrow$ crossover frequency ω_c but $\downarrow$ phase margin.